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Safety, Health & Environment Rules and Regulations

(Mandatory Contract Requirements)

R05.3- March ,2022

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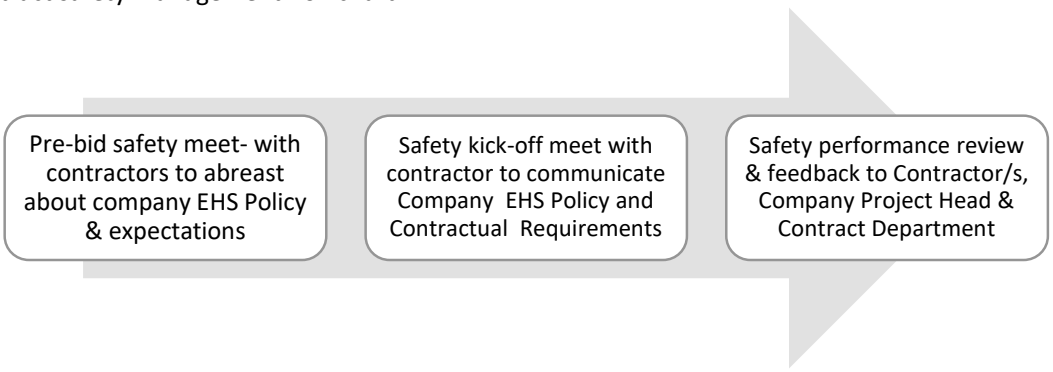
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1. Introduction

The Safety, Health & Environmental Protection requirements are business core values of the Employer and it is applicable to Oberoi Projects and its subsidiary companies here in after referred as "Company". These Rules & Regulations are mandatory requirements and abiding on employees, contractors, sub-contractors, vendors, consultants and visitors. The Company is confident that religiously adherence of these rules by the Contractor will ensure elimination of hazards, reduction in risk and prevention of incidents.

The Project Head, Construction Managers and Project Safety Manager of the Company has been empowered to enforce these rules and regulations, which will be strictly monitored through periodic inspections, measurements, and audits. Any breach of these rules and regulations will attract appropriate disciplinary action or penalty or both.

Contract safety management flowchart:



2. Applicability of Rules & Regulations

The prescribed Safety, Health & Environmental rules and regulations are mandatory for all category or package of Contractors, however certain requirements are not applicable to small package of Contractors and a separate guideline will be issued to them basis nature of work, hazard & risk, equipment used and manpower strength.

The Contractors should refer to the applicable clause and make sure that safety compliance is achieved and the objective evidence is available for verification and to the satisfaction of the Company Safety Manager, to prevent disciplinary or penalty action.

The Contractor shall be fully responsible for ensuring safety, health and environmentally safe working conditions related to their Works in the project premises including prevention of incidents and injury while performing contractual obligations. Any consequences as a result of Contractor's negligence (including but not limited to, accident, injury or death to its workers, legal proceedings, statutory issues and / or property damage, third party loss/damage etc), shall be dealt by the Contractor by himself and the Company shall be deemed to be indemnified and protected against any losses (including but not limited to, legal and monitory expenses) whatsoever as a result of the aforesaid negligence. The Employer shall have the authority to recover any losses incurred on behalf of the Contractor as a result of Contractor's negligence from the outstanding amounts due to the Contractor.

3. General Safety and Health Rules

3.1. The Contractor shall abide by the Company's Environment, Health & Safety Policy, EHS Golden Rules, Procedures and Guidelines. Ignorance to this and deliberate interference with critical safety system or placing others in danger is not acceptable and shall be dealt seriously.

3.2. Contractor must know their "EHS LEGAL" responsibilities under The Building & Other Construction Workers Act 1996 (BOCW Act) & The Maharashtra State Building & Other Construction Workers Rules 2007 (MSBOCW Rules) and other common law for their establishment, employees & neighbouring community/ property and shall conduct their business confirming to best engineering and administrative practices.

3.3. Safety induction and medical examination of new workers and staff is mandatory and shall submit record to the Company.

3.4. The inspection of tools, plant & machines for its fitness by the safety department is must. The contractor shall time-to-time notify the Project Head & Safety Manager of the Company about new deployment of manpower and machines.

3.5. During safety induction, trade skill of workers should also be verified and mention on their induction form and ID card.

3.6. The contractor shall provide & maintain all safety requirements prescribed in this document. All cost related to these requirements as directed by the Company but not limited to the items listed, shall be entirely borne by the Contractor till completion of the work/project.

3.7. The Contractor shall ensure that every worker, staff, sub-contractors, vendors, suppliers, visitors at site has been provided with mandatory safety PPE (safety helmet, safety shoes, safety reflective jacket, Safety harness, Safety goggles, ear-muffs) and other necessary PPE as per work & hazards and ensure that they shall wear it all the time.

3.8. Site access is subject to strict check by the Oberoi Security for PPE, ID Card & legal compliance, the Contractor shall issue photo ID Card by authorized signatory of Contractor.

3.9. Contractor shall perform work as per approved method statement and risk assessment only.

3.10. Consumption of or working under influence of any alcoholic liquor, illegal drug or other intoxicating substance is prohibited, any violation of rule shall attract eviction of such person from project premises

3.11. Possession or use of lighter, matchboxes, cigarettes, firearms, ammunition, fireworks, flammable chemicals etc prohibited, other than the one permitted for site work.

3.12. Smoking and tobacco chewing in the site premises and labour camp is prohibited, any violation of rule shall attract eviction of such person from project premises.

3.13. Fighting, bullying, harassment, horseplay or other violent actions or self-harm or attempted suicide prohibited.

3.14. Ignorance to instructions, rules and signage's and deliberately contravene of safety rules or interfere with critical safety system or placing another employee, contractor or visitor in danger shall be dealt seriously.

3.15. In the event of any emergency or fire broke-out, all personnel should report at nearest assembly point or safe location.

3.16. All height work (above 2m) shall be performed with provision of required safe means of access, working platform, lifeline and safety harness and safety net.

3.17. Electric Distribution Boards (DB) shall not be used unless provided with 30mA Earth Leakage Circuit Breaker (ELCB)/ RCCB and danger sign, insulation rubber mat, diagonal two point earthing of panel body and required sockets.

3.18. Contractors are fully responsible to supervise their work, machines and workers and must deploy competent supervisors.

3.19. Use of Bamboo scaffolding is prohibited at Company Projects.

3.20. Maintaining general housekeeping at site is basic responsibility of the Contractor. Debris and other scrap shall be removed on daily basis to designated location to ensure obstruction free safe working condition at site.
3.21. Dropping of debris and other objects from Towers or in void or shafts is prohibited and punishable by imposing heavy penalty.
3.22. Contractor shall ensure that all hand tools are secured by tying lanyard to prevent accidental drop from height.
3.23. Shall keep First Aid Box as per BOCW Requirement at site and office and display "First Aid" signage prominently.
3.24. Contractor shall submit TPI Certificate of lifting tools, tackles, slings, cranes etc., before use.
3.25. Contractor shall not remove existing fall protection barricades & safety nets without work permit and without arranging alternate protection.
3.26. Hot work permit for gas cutting, welding, brazing etc job is mandatory. Ensure trained workers, basic PPE, welders face shield, ABC type 6kg portable fire extinguisher, fire blanket & flash back-arrester on gas cutter-set. Shall handle and transfer gas cylinders using wheel mounted trolley only.
3.27. Contractor employees shall use available urinal, toilet and washing facility. Any violation of urinating /defecating at site premises is serious offence and punishable.
3.28. In the event of Contractor fail to maintain above safety, health and environment requirements, Company reserve the rights to ensure or restore safety conditions and back charge the Contractor for its time & cost.

4. Environment, Health & Safety Plan of Project (EHS Plan)

The Contractor has both legal and contractual responsibility to formulate and implement SMART EHS Plan before proceeding for any construction activity. EHS Plan shall be prepared as per Company guidelines and approval of Company Project Head is mandatory. The emergency management plan as per MBOCW (Rule 63), shall be part of the EHS Plan. The Contractor should submit EHS Plan within 15 days after receipt of the work order or letter of intent, whichever is earlier.

5. Method Statement, Hazard Identification & Risk Assessment and Safety Audit

i. Method Statement and HIRA: Contractor shall actively identify the job specific hazards to eliminate or reduce the risk to no harm level. The contractor shall submit method statement after incorporating identified risk and mitigation measures. The method statement and risk assessment must be approved by Company Safety Manager, Tower In-charge and Project Head. The contractor must adhere to the approved work method statement and any deliberate deviation shall not be acceptable and could lead to stoppage of work and monitory penalty.
ii. Safety Inspections: Periodic scheduled safety inspection of the workplace, construction activity, plant & machines, hand tools, lifting equipment, power tools, electric system, gondola, hoists, barricades, storages, safety nets, PPE etc shall be responsibility of the Contractor. The Contractor shall submit the monthly inspection schedule to Company Safety Manager.
iii. Safety Audit: Major contractors shall mandatorily conduct third part safety audit through external agency, which is approved by the Company on six months periodicity and submit report and compliance plan to the Company within 15days after audit.

6. Safety Officer and Safety Supervisor

The contractor is fully responsible for safety supervision of their construction activities, workers, plant & machineries, lifting tools & tackles, hand tools, scaffoldings & platforms, electrical system, labour camp, batching plant etc. To ensure work is performed safely and incidents are prevented,

the deployment of safety officer and safety supervisor is mandatory. The contractor shall appoint sufficient number of safety officers approved by the Company Project Head. The Contractor shall employ only qualified and competent safety resources as prescribed by BOCW Rules and by Company.

	Scope of Work and Safety Risk	**Safety Manager & Safety Officer**	**Safety Steward**
1	Civil work (shell & core) of Tower	One DGM/ Sr. Safety Manager as overall in-charge. In addition, one Safety Officer per Tower & for non-tower.	One Safety Steward per 100 workers or one stewards per 20 floors (or as per site condition decided by Company Project Head)
2	Major Contractors (Other than Civil): Excavation; Pilling; MEP works; Façade; Drywall; External painting; External putty; Structural steel work; lift installation work etc or Contractor employing 50 or more workers and involved in hazardous work (height work, hot work, confined space work, blasting etc)	One safety officer as in-charge	One safety steward per tower
4	Medium Size Contractors: Gypsum / Putty; False Ceiling; Acoustics; Railing work; Glass Partitions and Doors; Glass Railing; Interior works; Internal Painting etc or other work Contractor employing 100 or more workers involved in hazardous work such as height work, hot work, confined space work, excavation, demolition, piling, high-rise scaffold erecting/dismantling etc or as directed by company Project Head.		One safety steward per tower & non-tower

Safety Staff Competency Criteria: The Contractor shall employ only trained and experienced safety resources as per below criteria:

	Safety Resource Category	**Qualification**	**Experience**
1	DGM/ Sr Safety Manager/ Manager	Advance Diploma in Industrial Safety (one-year course from State Board of Technical Exam, recognized under State Factories Rules / BOCW Rules or Post Graduate Diploma in Construction Safety Management from NICMAR.	Minimum 8-15 years' experience in construction safety sector. Duties: as laid down in Schedule VI of MBOCW Rules.
2	Safety Officer	Advance Diploma in Industrial Safety (one-year course from State Board of Technical Exam, recognized under State Factories Rules / BOCW Rules or	Minimum 5-10 years' experience in construction safety sector.

		Post Graduate Diploma in Construction Safety Management from NICMAR.	Duties: as laid down in Schedule VI of MBOCW Rules.
3	Safety Steward / Supervisor	Graduate in any discipline and shall possess short term safety & fire management course from NIFE, NILEM, IOSH, Annamalai University etc	minimum 4-8 years of site safety experience and adequate knowledge of subject

7. Personal Protective Equipment

It is mandatory at Company Projects to wear Safety Helmet, Safety Shoes, Reflective Jackets and where required safety belt for height work. The Contractor shall be responsible to provide necessary PPE to their staff, vendor and workers at his own cost and must adhere to Rule 70, 72, 73, 81, 205 etc of MSBOCW Rules. All PPEs shall be of standard quality & capable to provide complete safety & health protection. The quality shall be checked and approved by Company Safety Manager. PPE shall conform to relevant IS Standard / EN Standard as detailed below.

	PPE Items	Standard and Specifications
1	Safety Helmet.	MOC: HDPE; IS:2925-1984 (Reaffirmed 2000) or EN Standard 397; chin strap & nap strap (adjustable); Inside- plastic head band. Sticker with name of contractor. -Staff white colour helmet; Safety staff- green colour helmet; Workers yellow colour helmet; Operators, mukadam & supervisor Blue Colour Helmet; Electrician red colour helmet. Visitor- White colour helmet with "Visitor" sticker.
2	Safety shoes (with steel toe-cap)	IS:15298-2002; EN20 345 certified by DGMS/CLI Lab; ankle height not less than 7cm, acid/alkali resistant sole; foam cushion in inner side for ankle lap.
3	Reflective safety jacket with fluorescent strips	Orange colour of suitable make and washable with Contractors company name/logo printed on it.
4	Full body industrial safety belt & harness	IS:3521-1999/EN 361; Standard lan-yard 1.8m; scaffold-hook and double lan-yard
5	Fall arrestor & Lifeline	Fall arrester: EN 353-2:2002 Lifeline: 12mm/14mm polyamide rope
6	Ear Muff / Ear Plug	IS: 6996-1973 (Reaffirmed 1998); IS 9167 or EN 352-1:2002 and EN 352-2:2002; made of sponge or foam; preferably with cord; length of cord not less than 50cm; after squeezing it should return to its original shape within 30 seconds.
7	Safety Goggles	IS: 5983-1980 (Reaffirmed 2002) or EN 166:2001; made of tough polycarbonate material; lenses with anti-scratch treatment; colour of lenses- Clear (UV clear).
8	Face shield	IS: 8521 (Part II) – 1977 (Reaffirmed 2002) IS: 8521 (Part I) – 1994 (Reaffirmed 2002) or EN 175F; IS-1179:1967.
9	Respirators/ Dust mask	IS: 15321 – 2003, IS: 15322 – 2003 and dust mask IS:9623
10	Hand Gloves (canvas)	IS 6994-Part-I; MOC-split or chrome leather; stitching should be firm; Cut-resistance proof
11	Hand Gloves (leather)	IS6994 Part-I; Moc- Coated fabric; additional lining at palm; firm stitching. Cut-resistance proof

12	Electricians hand gloves (rated for voltage)	IS-4770:1991-rubber gloves
13	Shoulder pads for reinforcement handling	Round neck type leather covered with foam cushion, as per manufacturer's specifications.
14	Gum Boots (with steel toe-cap)	IS-5557: 2004; IS 12254: 1993 (PVC)
15	Electricians' safety shoes (with non-conductive toe-cap)	As per manufacturers specifications and test certificate

a) Contractor shall maintain minimum 25% buffer stock of PPE at all time with respect to total man-power at site. In case contractor fail to provide PPE, Company reserve right to stop the work or provide PPE at cost & time of the contractor.
b) Contract shall handover PPE to staff and workers immediately after safety induction and shall demonstrate correct usage of PPE & safety belts.
c) It is mandatory that all supervisors, site engineers and visiting vendors should wear safety helmet, safety shoes and reflective jackets and use safety belt for all height work as required.
d) Workers employed on mixing asphaltic materials, concrete, cement and mortars shall use rubber gum boots with toe-cap & hand gloves and other PPE like goggles, respirators and hand gloves, etc.
e) Persons engaged in welding and gas cutting work shall use appropriate welding face shields, leather hand gloves and fire-retardant protective cloths. The persons who assist them shall also use appropriate eye protection goggles.
f) Workers engaged in rock breaking, grinding, drilling and chipping etc shall use protective eye goggles, dust respirators, dust mask, earmuffs /ear plugs, etc.
g) Persons working at height and exposed to risk of fall from height shall use full body harness double lan-yard safety belt with kinetic shock absorbers, fall grab-arrestor and lifelines ropes as per the risk.
h) **Confined Space Safety**: When workers are employed in sewers, septic tanks and inside man-holes which are in use, the contractor shall ensure that the Confined Space Entry Permit is taken, manholes are opened and adequately ventilated for sufficient time. After ventilation and before man entry, the atmosphere inside the space shall be checked for the presence of any explosive mixture, toxic gas or oxygen deficiency. The workers shall be allowed to get into the man-hole under safe working environment & supervision of watch-person only.
i) The man-holes opened shall be cordoned off with suitable railing and provided with warning signals or caution boards to prevent accidents. There shall be proper illumination. All safety measures for working in confined space as given in the BOCW shall be adhered. In case of forced ventilation, battery backup for ventilation and measures to rescue workers shall be ensured.

8. **Fire Prevention and Protection**

The contractor shall take all necessary precautions to prevent outbreak of fires at the construction site and at the same time keep adequate provisions to extinguish fires. Any fire risk and personal or property damages posed out of negligence or ignorance to rules by contractor shall be at cost and risk of respective contractor. Fire is most successfully fought when the extinguishing equipment is suitable and persons are trained. Following rules shall be adhered:

8.1. Fire extinguishers suitable for the different class of fires such as Class A, B, C & D as per IS: 2190-1992 (Reaffirmed in 2010) and sand buckets shall be made available at designated "Fire Points" at each floor and in podium & basements at every 15m with unobstructed access

marked by location signages. The fire protection layout shall be approved by the Company Safety Manager.

8.2. Impart training to employees in efficient firefighting operation with hands -on training and six-monthly refresher training.

8.3. The major Contractor shall furnish list of trained people in firefighting to company and also display list at site and office.

8.4. The contractor shall adhere to MSBOCW - Rule 62: Fire Protection and shall make all necessary provision for prevention & fire fighting.

8.5. As per Company Rules entire construction site is "No Smoking Zone" and must display warning instructions and signages prominently at conspicuous places. Any ignorance shall be treated as major violation and person/contractor will be barred from entering site. The Contractor shall communicate fire safety rules to staff & workers during safety induction.

8.6. All hot work at site like welding, gas cutting, brazing, heating etc shall be undertaken under Hot Work Permit & under supervision of contractor's safety officer.

8.7. Quantities of combustible/ flammable materials like timber, shuttering ply, coal, paints, diesel etc. shall be the minimum required in order to reduce the fire load to minimum.

8.8. Containers of flammable paints, thinners, diesel, grease and allied materials shall be stored in a separate room at ground level which shall be well ventilated and free from excessive heat, sparks, flame or direct rays of the sun. The containers of paint shall be always closed by lids. The store shall be clearly marked as Flammable Material and a fire point shall be located near such storage.

8.9. The Portable Fire Extinguishers, Fire Buckets, Fire Hose-Reels, Fire Water Tank, Fire Pump, etc shall be inspected once in a month for fitness conformity and log book shall be maintained. The date of last inspection and maintenance shall be displayed prominently on fire extinguisher body.

8.10. The Contractor shall conduct fire emergency drills and site evacuation on quarterly basis.

8.11. The emergency telephone number of the nearest fire station shall be displayed at suitable locations (near site office, main entrance of the site, first aid centre, stores, security etc.) in bold distinct font.

8.12. All hot work shall be backed by 6kg DCP/ABC Type portable extinguisher.

8.13. Fire-fighting equipment should be used for fire-fighting purposes only, never place obstructions within 15 ft. of fire extinguishers and hose houses.

8.14. The Civil Work Contractor shall install the fire hydrant line with riser and hose-reels located every floor backed by adequate fire water supply and pressure. The scheme should be approved by Company Project Head.

9. **Housekeeping**

9.1. Company value Good Housekeeping as foundation stone for good safety culture and management believe that many incidents can be avoided just by ensuring clean and orderly construction site. The contractor must adhere to Housekeeping Guidelines of Company.

9.2. The respective Contractor shall be responsible for upkeep of their working place and surrounding area and should not be obstruction & unsafe for others.

9.3. Housekeeping management shall be part of the Site EHS Plan and Site Logistics Plan.

9.4. The contractor shall follow the recommendation of IS: 4082-1996 (Reaffirmed in 2003) for stacking and storage of construction materials and components at site.

9.5. The plan of temporary structures shall be such that they do not hamper easy movement of worker, vehicles & cranes. No materials on any of the sites of work shall be so stacked or placed as to cause inconvenience to any person or the public.

9.6. The Company Project Head has the right to stop work if the Contractor fails to improve upon the housekeeping after having been notified.

9.7. The Company may instruct the contractor to remove any materials which are considered to be of danger or cause inconvenience to the public.
9.8. After completion of the work, the contractor shall remove all scaffoldings, surplus materials, scrap, rubbish and all temporary structures, huts and sanitary arrangements used/installed for his workmen at site.
9.9. The contractor shall stack all undesirable materials and debris to the designated scrap yard area at his own cost, as directed by the Company.
9.10. The major Contractor shall deploy dedicated & sufficient housekeeping team as per the size of the project but not less than 5% of total workforce and should be identified by reflective jackets as “housekeeping Team” (recommended “yellow” colour jacket with reflective bands), controlled by Responsible Person of the Contractor, who should schedule housekeeping of site, perimeter, internal access roads and all associated places such as offices, stores, workshops, stock yards, batching plants, labour camp etc and maintain a register.
9.11. The Civil Contractor shall ensure separate gates & access roads for labour entry and material vehicles to prevent workers directly coming into contact with vehicle traffic.
9.12. The walkways, gangways with minimum width of 1.2m shall be clear of debris, protruding dowels, timber with nails, oil spillages & other obstructions all the time and prevent tripping danger. Similarly all emergency exit, fire doors, firefighting equipment, access to electrical panels & DB’s, first aid stations, ambulance station, stairways, ladders, scaffolds etc should be unobstructed, dry and in good working order.
9.13. The Contractor shall ensure adequate illumination in the workplace and maintain general uniform illumination of minimum 50 Lux across length & breadth including staircases, passages etc.
9.14. The Contractor shall ensure that parking of trucks, transit mixers& other construction vehicles not stranded on public roads, which may impair aesthetics & obstruct traffic. The tires of the trucks leaving the site shall be cleaned with water, wherever the possibility of spillage on public roads, particularly during excavation, piling, rainy season etc.
9.15. Dewatering of stagnant water from site shall be regular activity equipped with adequate capacity of pumps.
9.16. It shall be daily routine that sufficient time is devoted on housekeeping of the workplace after completion of work & before leaving the workplace.
9.17. The contractor shall use waste bins for collection and segregation of waste like steel scrap, wood/ply, aluminium, empty drums, corrugated boxes, debris and other waste material etc and shall not be left scattered at workplace.
9.18. Similarly, on weekly basis or as required remove all the scraps, debris, waste oils and other disposables, to the designated yard. And same shall be disposed of from the site on monthly basis or depending upon the quantum of scrap.
9.19. Contractor shall arrange to clean safety nets, catch platforms & deposits on chajjas weekly.
9.20. Similarly, all surplus earth and debris shall be removed from the workplace to officially designated dump yard as per logistics plan fortnightly or as required.
9.21. The major Contractor shall install debris chute or other appropriate means in the buildings for safe removal of debris after approval of Project Head. Dropping of debris from tower or any height is strictly prohibited.
9.22. Over and above as a good practice the Contractors must plan at least a day in a week as a “housekeeping day” to keep the site orderly.
9.23. Electrical cables, welding leads shall be routed overhead to prevent trailing cables, tripping hazard & electric shock. Overhead cable routing height should be minimum 2.5m inside the buildings and minimum 6.1 m in open area supported by stable & insulated stands/MS-poles or otherwise ensure that DB’s and welding machines are placed near by job location.

9.24. Materials shall be stacked at least 2m away from leading edge of slabs, MEP/ Shaft openings, roof edges, excavations, or trenches to prevent fall of material to below level.

9.25. The stock yard shall be well laid out with easy access and material stored / stacked in an orderly and safe manner and must be segregated.

9.26. Steel, shuttering & other materials etc shall not be stacked more than 1.5m height and also concrete block and brick should be stored to a height not exceeding 1.8m at ground level; however, while stacking material at floor level or slab consider load bearing capacity of structures.

9.27. Piles stack of cement bag shall not be more than 10 bags in height unless such stack piles is stacked in suitable enclosure or otherwise adequately supported.

9.28. Sites having space constraints shall limit the material inventory or staggered inventory, disposed of scrap/ debris & other excess or waste material more frequently and plan rack stack methods.

9.29. Ensure that the work vicinity of hot work is cleared of all combustibles and rubbish and hot work permit procedure is strictly adhered.

9.30. The contractor shall display required nomenclatures and mandatory, cautionary & informative signage such as speed limit, no smoking, electricity danger sign, flammability diamond sign, no spitting, drinking water, urinal/ toilet with gender pictorial sign, empty & full gas cylinders storage, emergency exit, assembly point, numbering of floors/ levels etc as well as display caution signages for height work, falling objects, deep excavation, fire, electrocution, unauthorized entry or as per site requirement etc prominently.

10. Lifting and Hoisting Machinery

The use of lifting machines and tools/tackles including their attachments, anchorage and supports shall conform to the Rules 82 to 108 of chapter XI of MSBOCW Rules, 2007 and shall also conform to the following conditions.

10.1. Lifting machines and tackles shall be of good mechanical construction, sound material and adequate strength and free from any defects and shall be kept in good repair and in good working condition. Every rope used in hoisting or lowering materials or as the means of suspension shall be as per manufacturer's guidelines, of good quality and adequate strength and dimension and free from any defect. Test certificates of such ropes, slings, hook, D-shackles, etc. shall be submitted in advance to the Project Head.

10.2. The crane operator and other lifting appliance operator shall be properly trained and competent. No person under the age of 21 years shall be in charge of any hoisting machine or to give signal to operator of such machine.

10.3. In the case of every lifting machine (and of every chain, ring, hook, D-shackle, swivel, and pulley block used in hoisting or as means of suspension) the safe working load (SWL) shall be ascertained and clearly marked. In case of a lifting machine, having a variable safe working load, each safe working load and the conditions under which it is applicable shall be clearly indicated. No part of any machine or any gear shall be loaded beyond the safe working load except for the purpose of testing. This shall be approved by the Safety Manager of the contractor.

10.4. The base of such hoisting equipment shall be kept in perfect horizontal condition since any tilt would reduce the load carrying capacity of the equipment. The foundation shall be firm enough to support the equipment. The level shall be checked every day before starting the work in case of mobile hoisting equipment.

10.5. Double sling shall be used for hoisting material. The angle of the sling shall be wide enough for safe hoisting and the sling shall be adjusted as per the centre of gravity of the material to be lifted. A guide rope (manila rope of sufficient length, normally 1.5m long) shall be attached to the end of the material lifted in order to pull the same conveniently during lowering.

10.6. The Contractor shall notify the safe working load (SWL) of the lifting machine to the Company Project Head whenever new machinery brought at site and submit TPI certificate of Competent Person, without which machines shall not be taken into operations.
10.7. The contractor shall arrange thorough inspection and load testing of all lifting machines and tackles by the Competent Person, at least once in every six months and the records of such inspection and testing and updated inventory shall be maintained, and a copy shall be submitted to Company.
10.8. Motors, transmission, couplings, belts, chain drives and other moving parts of hoisting appliances shall be provided with adequate safeguards.
10.9. The contractor shall maintain a Register of Periodical Third Party Test (TPI) and maintenance history card of all Lifting machines, hoists and Gears at site as per Rule 97, Chapter XI, MSBOCW Rules and submit the same to Company Project Head periodically for examination.

11. Compressed Gas Cylinders

11.1. The Gas Cylinders together with their valves and other fittings and identification colours shall be maintained in good condition.
11.2. No cylinder shall be subjected to any heat treatment or exposed to a high temperature or to the sun or stored with flammable or explosive material.
11.3. Every cylinder containing compressed gas shall have its valve securely closed so as to prevent leakage. Valves fitted to the cylinders containing LPG and highly toxic gases shall be provided with security nut on the outlet to act as a secondary means of safeguard against leakage of gas.
11.4. If the leak in the valve cannot be rectified by tightening the gland-nut or the spindle, the cylinder shall be removed to an open space where it is least dangerous to life and property and the Supplier shall be informed.
11.5. Domestic LPG gas cylinders are not allowed to use at company site.
11.6. Cylinders shall be stored in upright position and adequately secured.
11.7. Gas cylinder handling shall be done by using wheel mounted trolleys or cradles and cylinders should not dragged or rolled. The cylinder set should remain on trolly while at workplace.
11.8. The cylinders shall be handled carefully and not be allowed to fall on one another or subjected to any undue shock.
11.9. Sliding, dropping, rolling or playing with cylinders is prohibited.
11.10. Cylinders shall be stored in cool, dry, well ventilated place under cover, away from open flames or any potential sources of heat.
11.11. The storage room or shed shall be of fire-resistant construction and well ventilated.
11.12. Liquified, Thin-walled cylinders such as LPG and dissolved gas (DA) shall not be stacked in horizontal position. All cylinders shall be stored in up-right position only.
11.13. Cylinders containing flammable and toxic gases shall be kept separated from each other by 3m and from cylinders containing other types of gases by an adequate distance or by suitable partition wall.
11.14. Cylinders shall not be stored with any combustible materials.
11.15. Empty cylinders shall be segregated from filled ones with signage and care shall be taken that the valves are tightly shut.
11.16. The contractor shall ensure that hydro-static test certificates of all pressurized gas cylinders are obtained & record available.
11.17. No lifting magnet shall be used in loading or unloading of cylinders filled with compressed gas. When any such operation is carried out by means of a crane or fork lift truck, a proper cradle of cylinder height should be used as approved by safety officer.

12. Diesel Storage Area

The Contractor intend to store Diesel or any other fuel shall adhere to the requirements of The Petroleum Act & Rules including following specifications:

12.1. Covered Storage area with PCC and sufficient slope towards collection pit. Provide bund of 110% of stored volume of fuel. Ensure adequate ventilation to prevent accumulation of vapours.
12.2. As far as possible, fuel drum shall be placed over drip tray.
12.3. No light fixture or flame proof light to be fixed inside the storage area.
12.4. No electric connection shall be installed inside the storage area.
12.5. Hand pump shall be used for fuelling vehicles or transfer from one container to another.
12.6. Foam type and DCP type extinguisher of adequate capacity shall be placed outside the storage area.
12.7. No Smoking, No Hot Work, No open Flame signage shall be prominently displayed outside.
12.8. MSDS and Immediate first aid measures shall be displayed.
12.9. For storage of Class-B petroleum i.e Diesel exceeding 2500liters, license shall be obtained.

13. **Electric Arc Welding and Gas Cutting (Hot work)**

13.1. All the hot work shall be carried out strictly after obtaining Hot Work Permit from the Company Site Engineer.
13.2. The contractor shall ensure site is cleared off all combustibles and use fire blankets to contain weld spatters and prevent fire.
13.3. For welding sets, all power connections shall be routed through ELCB of 30mA rating DB. Double earthing shall be ensured to the welding machine. A provision of a separate return lead shall be ensured connecting welding object and machine.
13.4. The cable to be used shall be of adequate capacity corresponding to output of the welding transformer / generator and shall be routed through dry isolated path. Welding cable terminals shall be provided with lugs and connected properly. Proper insulation of cable with insulation tape of approved quality shall be ensured and only double insulated cable shall be used. Extension of welding cables shall be done using standard connectors.
13.5. Personal contact with the electrode or other live parts of electric welding equipment shall be avoided. Wires and cables shall not be hung from any metal hook.
13.6. Accidental contact of electrodes with ground shall be prevented.
13.7. The welding cables shall not be allowed to get entangled with power cables. It shall be ensured that the cables are not damaged by movement of materials. Dragging and coiling of cable shall be avoided.
13.8. Barrier screens shall be erected to protect other persons from harmful rays and sparks from the work. When welding or gas cutting is carried out in elevated positions, precautions like providing metal sheet, fire blankets etc. shall be taken to prevent sparks or hot molten metal falling on persons or flammable/ combustible materials below
13.9. Adequate ventilation shall be provided for easy dispersion of toxic fumes while welding, brazing and cutting in confined space.
13.10. Suitable type of protective clothing consisting of fire resistant gauntlet gloves, boots and aprons shall be provided to workers to protect from heat and hot molten metal splashes. Welding shields with filter glasses of appropriate shade shall be worn as face protection against UV & IR rays.
13.11. Welding and gas cutting shall not be carried out by standing on drums, barrels, tanks or other containers.
13.12. One 6kg ABC type portable extinguisher shall be compulsorily stationed near welding/ gas cutting work.
13.13. Contractor shall ensure that the hot work location and surrounding is inspected by fire watch at least for half hour after the hot work is over.

14. **Electrical Safety**

I. Before commencement of construction work, the Contractor shall take adequate measures to prevent worker coming into physical contact with any electrical equipment or apparatus, machines or live electrical circuit which may cause electrical hazard.

II. The Contractor shall display and maintain suitable danger warning signs on all electric panels and DB's in Hindi or local language understood by the majority.

III. In construction work where the exact location of underground electric power line is not known, the building workers using jack hammers, crow bars and other hand-tools which may come in contact with a live electrical line, shall be provided with insulated tools, protective gloves and foot-wear.

IV. At excavation site, the Contractor before using excavators and other excavating equipment shall conduct survey to trace live cables and disconnect them.

v. The Contractor shall ensure that, no wiring, which may come in contact with water or which may damage mechanically is left on ground or floor.

vi. Contractor shall ensure that all temporary electrical installations are provided with earth-leakage circuit breakers (ELCB or RCCB of 30mA).

vii. The Contractor shall deploy licensed electrical staff (approved by Government licensing authority) at site and submit list to Company.

viii. The Contractor shall provide required PPE to electricians as per Indian Electricity Rules.

ix. It shall be the responsibility of the Contractor to provide and maintain complete installation on the load side of the supply point with regard to the safety & maintenance.

x. All cabling and installation shall comply with statutory requirements i.e. The Electricity Act, 2003; Indian Electricity Rules, 1956 (as amended in 2005); National Electric Code 1985 (as amended in 2005); Other relevant rules of Local Bodies and Electricity Boards

xi. The Contractor shall submit schematic and single line diagram for approval of Company.

xii. Method of earthing, installation and earth testing results shall conform to IS Specifications [IS: 3043 - 1987 (Reaffirmed in 2001)].

xiii. All three-phase equipment's MS cabinets and machine body shall be double earthed diagonally.

xiv. All light fixtures and portable equipment shall be effectively earthed to main earthing.

xv. All earth terminals shall be visible and earth pits shall be identified and numbered. Neutral conductor shall not be treated as earth wire.

xvi. The entire installation shall be subjected to the following tests before energizing of installation including portable equipment: Insulation resistance test, Polarity test of switches, Earth continuity test, Earth electrode resistance.

xvii. Electrical equipment and installations shall be installed and maintained as to prevent danger from contact with live conductors and to prevent fires originating from electrical causes like short circuits, overheating, etc.

xviii. The minimum clearance to be maintained from HT overhead lines shall be 6.10m (minimum) as per the Rules 77-80 of Indian Electricity Rules, 1956 (Amended in 2005).

xix. No material or earth work shall be allowed to be dumped below or in the vicinity of the bare overhead line conductors, which may hamper minimum safety clearance.

xx. Grounding conductor of wiring system shall be of copper or other corrosion resistant material. A double grounding connection shall be made in appliances/equipment where chance of electric shock is high.

xxi. Electric fuses and/or circuit breakers installed in equipment circuits for short circuit protection shall be of proper rating.

xxii. Wires and cables shall be adequately supported and an approved method of fixing shall be adopted. Loose hanging wires & cables shall be avoided. Lighting and power circuits shall be kept distinct and separate.

xxiii. Reinforcement rods, props, scaffoldings or any construction metallic part of structure, trees trunk or branches shall not be used as support for wires and cables, fixtures, equipment, earthing, etc.

xxiv. Overhead cables shall be routed minimum 6 m height on insulated MS poles/ stands and all cables inside buildings shall be routed 2.5m height to prevent trailing, tripping and electrocution. Floors shall be free of trailing electrical cables to avoid tripping hazard.

xxv. The Site Distribution Boards and extension boards shall be as per IP-55 specification i.e. dust & water ingress proof mounted on stand or wall and should have individual switch for each socket.

xxvi. Industrial type three-pin plug (iron clad) only shall be used. Open /naked wire connections are prohibited.

xxvii. Open type and wooden frame DBs are prohibited to use at company projects.

xxviii. Isolating switches shall be provided close to equipment for easy disconnection of electrical equipment or conductors from the source of supply in emergency and repair.

xxix. In front of distribution boards (DBs) a clear space of 1.0 m shall be maintained as access during an emergency.

xxx. All electrical equipment to be used in explosive environment shall be explosion proof.

xxxi. All connections to lighting fixtures, starters or other power supplies shall be provided with PVC insulated, PVC sheathed twin/three/four core wires to have better mechanical protection for preventing possible damage to equipment or injury to personnel. Taped joints shall not be allowed.

xxxii. All the lighting fixtures and lamp holders shall be of good quality and in good condition.

xxxiii. Only PVC insulated and PVC sheathed wires or armoured PVC insulated and sheathed cables shall be used for external power supply connections of temporary nature. Taped joints in the wires shall not be used.

xxxiv. Lamps used for illumination and testing purpose shall have cover or guard to protect them from accidental breakages. Only 24 Volt supply system shall be used for hand lamps etc, while working inside metallic tanks, vessels or confined space.

xxxv. As per provision of Rule 47-A of IE Rule 1956, license to operate the DG shall be mandate to contractor.

xxxvi. First Aid treatment chart of electrical shock shall be displayed at First Aid Centre.

xxxvii. While working on or near a circuit, whenever possible the use of one hand may be practiced even though the circuit is supposed to be dead. The other hand may preferably be kept in pocket.

xxxviii. Operation of electrical equipment and welding equipment shall be avoided when standing on wet floor or when hands are wet.

xxxix. While tapping electricity from the socket, three-pin plug top must be used. It shall be ensured that no extension boards are over loaded while tapping.

xl. Use of domestic two core cables and PVC extension boards are prohibited.

xli. Power supply to machines and lighting fixtures shall be switched off when not in use.

xlii. Temporary electrical connections shall be removed as soon as the stipulated work is over. After completion of the works, the contractor shall dismantle the distribution boards and the other facilities he may have erected.

xliii. Electrical Safety work permits & LOTO shall be used for switching off the main feeder and equipment by the contractor during batching plant maintenance, pan-mixer cleaning, tower crane & other machines maintenance.

xliv. Power tapping for electrical hand tools shall as near as possible to prevent trailing cables.

xlv. Inspection & testing: ELCB/RCCB shall be tested fortnightly, earth pits & earthing connections shall be tested quarterly basis and record shall be evident.

15. Portable Electrical Hand Tools & Equipment's

i. Portable electrical equipment shall be weekly examined, tested and maintained to ensure that the equipment and its leads are in good order. Log sheet shall be maintained for inspection recording the inspection dates & faults noted. The insulation and winding resistance of the portable electrical equipment shall be checked at least once in a month and report shall be submitted for all such machines.
ii. All portable appliances shall be provided with three core double insulated cables and three pin industrial plugs. It shall be ensured that the metal part of the equipment shall be effectively earthed.
iii. Connections to portable equipment or machines from the panel/distribution board/extension board shall be taken using 3 core double insulated PVC flexible copper wire in one length. No joints shall be allowed in this flexible wire. In case, single length of wire is not sufficient for a location then the supply can be tapped by providing another extension board comprising of switch and socket.

16. Hand Tools Safety

i. Impact Tools: The contractor shall use precision grip for the most commonly used impact tool, hammer for light work. For safe operations, the hammer shall have a straight cylindrical handle of 24 to 40 mm caliber with a maximum length of 600 mm and maximum head weight of 6.5 to 7.5 kg. Hammers shall be maintained such that cracked or weak handles are replaced and heads are in good condition and firmly secured to an undamaged handle.
ii. Cutting Tools: The contractor shall ensure that various cutting tools like axes, chisels and shovels, etc. are made up of material with adequate strength. The contractor shall ensure that wooden handles are to be moist before use during summer. Proper PPEs like hand gloves, ear plugs, goggles, dust respirators, etc. shall be provided to the worker as per the need of the work.
iii. All hand tools shall be secured by tying lanyard to prevent accidental drop from height.

17. Machine and Hand Tools Guarding

17.1 The Contractor shall ensure that rotating, reciprocating & dangerous parts of machines and hand tools securely guarded and they are never removed while machine is in motion and checked weekly by Contractor.

17.2. On completion of maintenance, oiling/ greasing any guards that was removed must be replaced, before machine again taken into operation.

17.3. Construction of guards shall be as per IS 9474, which shall prevent human body access to the rotating parts of machineries.

17.4. Grinding wheels of prescribed diameter and IS quality only shall be used, in order not to exceed the prescribed peripheral speed. Safety goggles shall be worn during grinding operation. Adhere Rule 64 MBOCW Rules.

18. Fall Prevention and Protection

Fall prevention measures shall be the foremost priority of the Company. The Contractor shall make sure all efforts to achieve this objective and ensure robust proactive and positive measures. The use of full body harness safety belt is minimum necessity; however it is the last level of protection and is not the substitute to the hard barricades, railings, safe working platforms, toe-guards, access staircase, ramps, safety nets and lifelines etc.

As a Mandatory Rule, the Contract shall not remove any barricades, railings & safety nets without the permission of Company Safety Manager and any deviation which many endanger life of people, shall attract stoppage of work & heavy penalty.

18.1. Barricades and Railings Standards

Basis building floor designs the contractor shall identify and prepare fall protection plan and get it approved from the Company Project Head. The excel tracker of fall protection plan prepared in tandem with Construction Planning Schedule, of horizontal & vertical openings such as MEP cut-outs, other services shafts, staircases, voids, lift shafts, windows, slab edges would be handy plan for Contractor for fabrication schedule. The protection fixtures shall be ready before shuttering work of the floor is initiated. The contractor shall ensure that railings & barricades shall be of following specifications and must meet BOCW & IS requirements.

	Fall Protection Requirements	**Specifications & MOC for Safety Barricades, MS jalli and Railings**
1	Leading edges of floors, top working floor leading edges, staircases; voids, MEP cut-outs, services shafts cut-outs, lift shafts, open/extended balconies, temporary platforms, scaffold railings, suspended platforms, elevated platforms, projected decks etc, connecting walk-way beams, refuge floors slab edges etc	MOC-40NB MS pipes, scaffold clamps & baseplates or 10mm MSGI wire rope, rated D-clamps, anchor fasteners and vertical monofilament/ PVC net cover: • Top horizontal guardrail at 1150mm height • Intermediate guardrail at 550mm • Vertical posts at not more than 3m apart. • Vertical Posts: Base plate 150 x 150 mm, rigidly fixed on floor by anchor fasteners. • Connector for railing pipes: scaffold clamps/ nut-bolts/ welding. • Toe-board of 150mm height. • Strength of railings: Capable to bear lateral load of about 100kg. • Neatly painted in red & white bands
2	Lift shafts and other services shafts Protection	Face side of shafts shall be closed by MS Grill: MOC- MS grill frame made of 35mm MS angle or 16mm steel reinforcement, inside mesh of 12mm rebar, spacing not more than 4" or single braded fencing MS mesh 4mm , properly welded & capable to hold lateral load of 100kgf. • Height of MS grill 1.80m minimum • Width of MS grill: Based on the shaft opening with 6" more at both side for fastening. • Lock-key arrangement and removal shall be by permit. • All joints welded and checked. • Fixing by anchor fasteners. • Entire opening shall be fully closed. • Removal of grills by Permit authorised by Project Manager of Contractor and Safety Manager of Company. • Contractor may use waste steel. Horizontal closing of Shafts at Intervals to Arrest Free Fall: MOC: MS grill frame made of 35mm MS angle or 16mm steel reinforcement, inside mesh of 12mm rebar, spacing not more than 4" properly welded, fixed side to side without gaps. • capable to hold vertical load of 1000kgf. • Slope with 30 degree towards landing. • Platform after every 2 floors to arrest free fall of person/ material through the shaft.

		• As a rule all shaft openings at shuttering/ working floor level shall be positively fully decked by MS platform, which will be climbing type.
3	Excavation and Pilling Work Fall Protection Railing	MOC-40NB MS pipes, scaffold clamps & baseplates: • Top horizontal guardrail at 1150mm height • Intermediate guardrail at 550mm • Vertical posts at not more than 3m apart. • For vertical posts: Base plate 150 x 150 mm, rigidly fixed on floor by anchor fasteners or grouted or 600 mm grouting in earth. • Connector for railing pipes: scaffold clamps/ nut-bolts/ welding. • Toe-board of 150mm height. • Strength of railings: Capable to bear lateral load of about 100kg. • Neatly painted in red & white bands • For open excavations railings shall be fixed 1m away from the edge of excavation. • Other shallow excavations shall also be barricaded by similar railing protection to prevent fall of person or vehicle.
4	Signages and visibility Colour	• For high visibility, the railings and barricades shall be coated by paint in red & white or wrapped with danger barricade tapes of same colour. • Place "Danger" sign board of 150 x 150mm
5	Mandatory: Contractor shall submit fall protection plan, tracker in excel sheet, scheme of barricades & safety screen and fabrication schedule for approval of the Company Project Head.	

18.2. Safety Wind-Screen and Safety Net Protection

The Civil contractor during RCC & related work shall provide safety nets protection with combination of vertical screens and horizontal catch nets as per building height & design to control the risk of fall of material or person as per following mandatory structured requirements.

The Contractor shall submit fall protection plan of safety netting for approval of Company Project Head.

1	Fall protection for Tower at (i) shuttering floors, de-shuttering floors and re-prop floors (top three operating floors) (ii) Void openings of tower and (iii) link beams of the tower.	Perimeter vertical safety wind-screen installation, made of MS frame of adequate strength and perforated MS screen sheets, climbing type hydraulically operated, covering all three floors, flushed with peripheral working platforms. Similar protection with same wind-screen shall be installed for voids and link-beams. The wind-screen should support till completion of the project cycle and should be installed at early stage of the project. The scheme and installation schedule

		shall be approved by company Project Head and Safety Head.
2	Protection below operating floors:	Safety catch nets- fan deck type 10m x 5m or suitable size just below operating floor, not more than 6m.
3	Protection at intermediate level along the building height	Safety catch nets- fan deck type 10m x5m or suitable size at interval of every 5 floors of tower
4	Peripheral Protection for ground level	All along the periphery of Tower extending 10m from face of building and erected at height not more than 5m above base of the building or at entrance slab level.
5	Prevent fall of objects below level- Net protection	All railings towards the external face of tower shall be vertically flushed by monofilament nettings.
6	Voids/ OTS/ Shafts	Horizontal safety catch nets across length & breadth flushed with wall to wall at alternate three floors.
7	Protection for working in external face of tower at height, external shafts, plumbing shafts, projected scaffolds/ platforms, projected/ open balconies, slab edge of floors, landings of open stairs etc and other high risk work at height.	vertical safety netting should be effectively used to envelope vertically the vulnerable work fronts to contain fall of material or person.

i. Horizontal catch nets shall be erected at an angle of not more than 20^0 to its horizontal sloping into the building and secured with building structure or MS props but shall not be tied with any temporary structure like scaffold, shuttering, temporary railing etc.
ii. In addition to hard barication, wherever required or in case where hard protection is not practically possible, vertical safety nets shall be effectively used to envelope the vulnerable work fronts to contain fall of material or person from the building. (Rule 68 & 206 MBOCW).
iii. Vertical Enveloping of Towers: The Contractor shall provide a standard safety net across all external faces of the Tower till the completion of the Project. This shall comply with Rule 41(3) of BOCW for all high-rise towers which are close to residential structures, commercial structures, public roads, internal roads, etc. The Contractor shall submit the methodology for anchoring and bracing of the safety net for the Employer's approval. The Contractor shall further ensure that the safety net does not obstruct the other activities of the Employer such as MEP works, façade works, finishing works, etc. The safety net shall be fire resistant and shall be able to withstand wind pressure for the Project. Charges for providing and installing the safety net is deemed to be included in the Contract Amount.
iv. The Contractor shall submit the safety net scheme to Company Project Manager and Company Safety Manager for his approval. Specification of safety nets shall be as per IS-11057:1980 or as prescribed by the Company.
v. Safety nets and support installations shall be drop-tested, after initial installation & before use, whenever relocated, after major repair and at 6-months interval if left in one place.

18.3. Lifelines & Fall-grab Arresters

While working with suspended platforms, gondola, mast-climber platforms, cantilever platforms, scaffolds, tower crane access ladder, work in lift-shafts & other shafts, external face of building etc the Contractor shall mandatorily ensure that the independent lifeline of 12mm/

14mm Polyamide rope of approved quality and fall-grab arrester is provided. Ensure that the lifelines are independently fixed to the building structure or any other rigid support and not tied with the equipment/ platform on which the person is working.

18.4. Working Platforms, Ladders and Scaffold

18.4.1. **Platforms**: The Contractor shall ensure that proper working platform are provided for finishing work on floor level and MCA area. The platform shall be light weight modular aluminium mobile scaffolds with adjustable platform with top & mid rail and toe-boards, heavy duty caster wheels with lock, access ladder etc with minimum loading capacity of 200 kg/m3 with required out-riggers for stability (to prevent toppling). The platform scaffolds shall be periodically inspected for fitness and tagged. The A-Type ladders being unstable and unsafe shall not be allowed at site, instead ladder with platform and safety railing shall be used.

18.4.2. **Scaffolding**:

The Contractor shall provide scaffolding of following minimum specifications:

- Heavy-duty scaffolding shall be used.
- Scaffolding of four meters above height shall have tie support with permanent structure at interval of 4 meters.
- Contractor shall deploy competent scaffolders for erecting and dismantling of scaffold.
- Contractor shall deploy certified scaffolding foremen, who will be responsible to inspect and provide fit to use-tag (Scaff-tag).
- Ladder shall be used for scaffold access.
- Working platform of scaffold shall be fully covered by using transom and metal plank, or hook type metal boards. Challi platform is not recommended, which is vulnerable to stability.
- Top hand-rails, mid rails and toe boards shall be fixed on scaffolding platform as per standard.
- Scaffold shall be erected over firm and plain surface. In case ground surface is not levelled than sole plate shall be used after levelling.
- Scaffold shall be erected over base plate over firm levelled surface or over sole plate having width thrice than base plate.
- Scaffolding erected above 2m height shall be as per approved scheme.
- Defective scaffold like bend, cracked, rusted etc shall not be used.
- Cross bracing shall be ensured for all scaffolds and for height above four jumps, diagonal bracings shall be given.

19. Incident Reporting and Investigation

The contractor shall report all type of incidents to the Company Project Head and adhere to reporting timeline as directed below. The FIR report should be followed by investigation report and corrective/preventive plan. Hiding or non-reporting of incident shall treat seriously and will attract heavy penalty and suspension of work. In the event of unfortunate incident & injury, it should be contractor's prime responsibility to provide immediate medical assistance to the victim of incident, any such delay or avoidance of responsibility by the contractor shall be dealt strictly.

Type of incident	Timeline for Report Submission

	Incident Reporting	Investigation Report Submission
First Intimation Report (FIR) for all type of incidents	Within 24 hours	Within 72 hours
Fire incident and dangerous occurrence	Within 24 hours	Within 72 hours
Loss time incident (LTI)	Immediate, and detail report within 72 hours	Within 72 hours
Fatality/ disability	Immediate, within 2 hours	Within 48 hours

20. Safety Induction of Employees

It is mandatory that the contractor shall impart effective induction to all employees including staff and workers before reporting at workplace. The major contractor shall have dedicated safety induction room equipped with audio-visual, safety film for screening, display safety policy, safety rules and instructions in language understood by workers. The safety officer of contractor shall be responsible to impart training, which should include explaining general safety rules of Company, probable construction hazards, risk, emergency & medical facility, fire protection & prevention, fall protection measures, electrical safety, safe rigging practices, use of PPEs, fire extinguishers, safety belts, safety nets, assembly point etc. The duration of safety induction shall be minimum two hours and feedback of employees is mandatory.

21. Safety Pep Talk / Tool-box Training

Due to literacy issue and social background of labours, daily sensitisation of the workers for safety & health adherence is mandatory. The Contractor shall conduct safety pep talks /tool box training of workers on daily basis in groups or tower-wise or activity wise. A mass TBT once in a week is must. The pep talks should cover work related topics like hazard & risk, use of PPEs and tools, housekeeping, hot work, electrical safety, use of safety harness for height work, health & hygiene etc. TBT shall solicit active participation of workers and site engineers and record shall be furnished to company on weekly basis.

22. Safety Awareness and Signboards, Posters

a. The Contractor shall at his cost conduct safety motivational and awareness campaigns for staff and workers and motivate them suitably.
b. The contractor shall display adequate numbers of signboards and posters prominently of safety awareness, caution & danger signs and informative signages as per site hazard and risk. They should be legible, prominently and placed at conspicuous places in the language understood by majority of the workers.
c. He shall have to maintain these signboards and posters in good condition throughout the contract period and shall have to replace them periodically.
d. Put-up Safety Park with display of all PPE and donning instructions.
e. Company deserve the right to direct contractor for adequacy of posters, instruction boards and signages (example: mandatory PPE pictorials, No Smoking signs, Flammable danger sign near fuel storages, DG sets, Caution for moving parts of equipment viz. fly wheels of piling rigs, motors of pumps, Scaffolding scaff-tags, Danger sign- deep excavation, floor openings, electrical danger sign, vehicle speed limit, fire extinguishers, exit signs, assembly sign, debris falling caution, height work & safety belt etc.

23. Ambient Air & Noise Pollution Monitoring

The Contractor shall conduct ambient air & noise monitoring survey through authorized agency at initial stage of the project and later every six months as per State Pollution Control Board / CPCB Standards and ensure necessary arrangement for control/ elimination of

pollution and suppression of dust by spraying water on dusty roads and affected areas. The noise pollution shall be controlled by effective engineering control. The Contractor shall adhere with Rule 61, 67, 70, 79 of MBOCW Rules. Shall submit copy of report to Company.

24. Drinking Water

The contractor shall provide adequate drinking water facility, which includes sufficient water taps, water purifiers and water coolers their staff and workers at his own cost.

a. Drinking water tanks should be cleaned fortnightly or as required and potability test should be done six monthly (as per IS:10500) and should be equipped with appropriate filtration media.
b. Also ensure that water tank lids are closed or follow directives of local Municipal Corporation, to prevent the larva growth and mosquito breeding etc.
c. Signage with pictorial shall be placed for drinking water in English, Hindi & local language.

25. Crèche : The Contractor shall provide crèche at site, if more than 50 female building workers are employed and it shall be under the charge of women trained in the care of children & infants and maintain in sanitary condition and adhere with Section 35 of BOCW Act.

26. Canteen : The contractor shall provide and maintain wherein more than 250 workers are employed, a canteen for the use of the workers and adhere with section 37 of BOCW Act.

a. Canteen shall be situated at the distance not less than 15.2 meter away from any latrine/urinal or any source of dust, smoke or fumes and should be equipped with fly catcher. Also waste water from canteen shall be carried away in covered drain.
b. Arrangements shall be made for the collection & disposal of canteen & labour camp food waste daily.
c. Medical examination of canteen food handlers shall be done twice in a year (if applicable).
d. At site provide & identify lunch shed with drinking water facility& dust bin. Eating food in work place is not allowed.

27. Shelter : The Contractor shall arrange suitable weather shelters for lunch and resting of the workers, at a safe distance away from construction activities.

28. Labour Toilet and Urinals : The Housekeeping and Hygiene at Company Projects is of utmost importance. The contractor shall ensure sanitation and hygiene at work place and at Labour Camp. He shall submit the sanitation logistics plan for site and camp for approval of Project Head of Company.

The latrines and urinals, as the case, shall be provided as specified below:

URINALS	**For Male Workers**	**For Female Workers**
1-50 workers	1 Urinal	1 Water-Closet per 50 female workers with separate entry & pictorial sign.
50-500 Workers	1 Urinal per 50 workers	
500 & above	Addition to above 1 additional urinal per 100 workers	
Inside the Towers by Civil Contractor	1 Urinal in the Tower after every two floors	-
LATRINS		
1-25 Workers	1 latrine	

25-100 workers	Additional to above 1 latrine per 25 workers	1 latrine per 25 female workers with separate entry and pictorial sign & notice
100 & above	Additional to above 1 latrine per 100 workers	

a. Every latrine shall be under cover partitioned off to secure privacy with proper door and fastenings, adequately lighted, and maintained in a clean and sanitation condition at all times.
b. Where both male and female building workers are employed, display outside each block of latrines or urinals a sign as "For Men Only" or "For Women Only" with pictorial sign of man or of a woman, as the case may be, written in the language understood by the majority.
c. The contractor shall employ full time sufficient sweepers to maintain latrins and urinal blocks in clean and hygiene condition and maintain logbook.
d. It shall be cleaned and disinfected at least every day and maintained properly by the contractor throughout the project duration.
e. Proper disposal of excreta by septic tank and soak pit shall be made by the contractor. In no case, the excreta shall be disposed off in any open drain, nallah, etc. which may cause outbreak of disease or reduce the overall hygiene of the workplace.
f. The floor area should be of impervious material easy for cleaning & sweeping.
g. Provision of sufficient illumination & pest control larva treatment & mosquito fogging shall be weekly.
h. White-wash of every latrines/urinals once in every period of four months.

29. Health Management and Medical Facility

The major contractor employing more than 500 workers at any given date of project schedule shall arrange adequate facilities for medical aid and treatment for his staff and workers including the first-aid facilities at the project site. In general the contractor shall adhere to the Rule 250, 253, 254, 255, 256, 258, 259 and Schedule III of MBOCW Rules, 2007.

29.1. **First Aid Centre**: The contractor shall ensure first aid centre (ambulance room) at site as per Rule 253 MBOCW Rules in order to facilitate immediate relief to the injured person. The centre shall be under charge of qualified Nurse, who shall maintain first aid injury & treatment record. The centre should be equipped with the articles specified in Schedule IX of MBOCW Rules. The size of the room shall be adequate for proper treatment of the injured persons and keeping the enlisted articles in an organized manner. The room shall be well ventilated and illuminated, preferably by natural means. The contractor shall keep a refrigerator of approx. 150 litres capacity for proper storage of injections and temperature sensitive medicines.

29.2. **Preliminary Medical Examination of Workers:** The Contractors shall at his cost conduct preliminary medical examination of workers before deploying at site and thereafter once in a year and submit fitness certificate to Company and ensure that the person having defective vision, deafness & has a tendency to giddiness are prohibited from employment. For height work, persons should be subjected to vertigo test and records should be available.

29.3. **Medical Examination of Operating Staff**: The Contractor shall arrange mandatory medical examination of all plant & machinery operators, vehicles drivers, crane operators & signaller, gondola & hoist operators, electricians and technicians before deploying at work, which shall include general fitness, mental alertness, vision, hearing, breathing tests and certificates shall be submitted to company in Form XXVIII.

29.4. **Medical Examination Record:** The contractor shall keep record of medical examination of every building worker employed by him in a health register in Form XXVIII MBCOW till end of project.

29.5. **Hospital Tie-Up for Medical Emergency:** The Contractors individually or collectively shall have official tie-up with the nearest well-equipped hospital for handling medical or accidental emergency at site.

29.6. **Ambulance:** Ambulance should be available any time at site or on call for emergency transportation. Telephone number & address of such hospital, doctor, ambulance services shall be prominently displayed. A copy of agreement with hospital shall be furnished to Company.

29.7. **Medical Officer:** The major contractor shall engage a medical officer as per Rule 256 MBOCW to provide medical consultation & treatment to the workers. The doctor's visiting timing shall be displayed prominently.

29.8. **Stretcher:** The Contractor shall keep minimum two stretchers in ready to use condition.

29.9. **First Aid Boxes:** The Contractor employing more than 10 workers shall have first-aid box available under the charge of trained first aider and put-up signage as "First Aid Box Available". The contents of first aid box should be as per Schedule XII of MBCOW.

30. Passenger cum-Material Hoists

30.1. The contractor installing the passenger-cum-material hoists for their use or the vendor providing such hoists for Company use on hire basis shall ensure comprehensive third-party tests & examination of the hoists by the Competent Person before being taken into use after installation or every shifting and there after thorough examination every six month and submit test certificate to the Company Project Head. Unless this is done the hoists shall not be used.

30.2. The Hoists case shall be so constructed as to prevent, when its gates are shut, any building worker carried by such hoist from falling out of it or from being trapped between any part of such cage and any fixed structure or other moving part of such hoist or from being struck by articles or materials falling down the hoist-way on which such hoist is moving.

30.3. The Hoists case shall be fitted on each of its side from which, access is provided to a landing place with a gate which has efficient interlocking or other devices to secure so that such gate cannot be opened except when such cage is at a landing place and that such cage cannot be moved away from any such place until such gate is closed.

30.4. Every gate at landings and hoist-way enclosure of such hoist is fitted with efficient interlocking or lock & key or other devices to secure so that gate cannot be opened except when the cage of hoists is at the landing place, and that such large cage cannot be moved away from the landing place until such gate is closed.

30.5. In every hoist used for carrying building workers there are provided suitable and efficient automatic devices to ensure that the cage of such hoist comes to rest at a point above the lowest point to which such cage may travel.

30.6. The approach to a hoist is adequately lit, fire extinguisher provided in hoist cage, safety operations instructions are displayed, and emergency ladder is available.

30.7. Display SWL prominently in the cage. Display maximum number of persons to be carried and also combination of material and person to be carried.

30.8. No person other than the operator trained, capable and medically fit to operate shall be employed to operate hoists.

30.9. The Contractor or vendor shall ensure that the preventive maintenance of the hoists is carried out as per manufacturers manual & frequency and daily inspection is conducted by operator before using the hoists and record available.

30.10. Floor landings along the hoist way shall be distinctively marked and legible.

30.11. Requirement of Rule No 92, 97, 100, 101 & 105 MBOCW Rules 2007 shall be complied.

31. Work Permit

All work having high risk of safety should be undertaken under proper "permit to work". Following work permit should be compulsory to be used at Company construction site.

i. Hot work permit: Gas cutting, welding, brazing, heating, etc.
ii. Height work permit: All type of work above 2m height from the floor level.
iii. Confined space work permit: Working inside silo, tank, manholes & pipelines/pits or any other location where there is a risk of insufficient air circulation or restricted entry space, etc
iv. Safety facility removal permit: Removal of any type of safety protection facilities including barricades, MEP covers, safety nets, lifeline/wire ropes barrication, periphery safety nets, safety screens, related signages, etc
v. Electric work: Working on electric supply system, panels, heavy machines operated on electric supply, batching plant etc
vi. Night work; Any work carried out in the evening beyond normal working hours of the site
vii. Excavation work: Any excavation work, trenching, cutting of earth beyond 2m deep

The permit shall be authorized by the Company Site Engineer/location incharge as per the standard procedure. The Company safety staff are authorized to verify the permit compliance and in case of violation/deviation, should stop the job till desired rectification is achieved.

32. Legal Compliance and Register

The contractor shall identify the requirements of good practices at site for fulfilment of legal requirement related to environment, occupational health and safety and welfare. The contractor shall enlist all the activities under the contract in advance in Legal Register and their effect on safety, health and environment. The contactor shall establish the procedure to take all legal approvals and timely renewal in advance. The contractor shall maintain an update Legal Register and Company Safety Manager shall verify quarterly.

i. Building & Other Construction Workers (Regulation of Employment & Conditions of Service) Act, 1996 and Maharashtra BOCW Rules 2007.
ii. The Gas Cylinders Rules 2016
iii. The Indian Electricity Act 2003
iv. The Indian Electricity Rules 1956 & 2005
v. Fatal Accidents Act, 1855
vi. Explosives Act, 1884 and Explosives Substances Act, 1908 & Explosives Rules, 2008
vii. Motor Vehicles Act, 1988 and Central Motor Vehicles Rules, 1989
viii. Petroleum Act, 1934 and Rules, 2002
ix. The Static & Mobile Pressure Vessels [Unfired] Rules 2016
x. Noise Pollution Regulation & Control Rules, 2000
xi. Environment (Protection) Act, 1986 and Rules, 1986.
xii. The Hazardous Waste (Management & Handling) Rules, 1989
xiii. Air (Prevention & Control of Pollution) Act, 1981 and Rules, 1982
xiv. National Ambient Air Quality Standards- Notification 1994 under Air Act.
xv. Water (Prevention & Control of Pollution) Act, 1974 and Rules, 1975
xvi. Battery (Management and Handling) Rules, 2001
xvii. Bio-Medical Waste (Management & Handling) Rules1998
xviii. Child Labour (Prohibition & Regulation) Act, 1986 and Rules, 1988
xix. Contract Labour (Regulation & Abolition) Act, 1970 and Rules, 1971
xx. Inter-State Migrant Workmen (Regulation of Employment & Conditions or Service) Act, 1979 and Rules, 1980
xxi. Public Liability Insurance Act, 1991 and Rules

xxii. Workmen's Compensation Act, 1923 and State Rules.
xxiii. Minimum Wages Act, 1948;
xxiv. Payment of Wages Act, 1936;
xxv. Employees Provident Fund and Miscellaneous Provisions Act, 1952;
xxvi. Employees State Insurance Act, 1948;
xxvii. Employee's Deposit Linked Insurance Scheme, 1976;
xxviii. Bombay Labour Welfare Fund Act, 1953;
xxix. Maharashtra Contract Labour (Regulation and Abolition) Rules, 1971;
xxx. Payment of Bonus Act, 1965;
xxxi. Maternity Benefit Act, 1961;
xxxii. Payment of Gratuity Act, 1972;
xxxiii. Compliances list as per Annexure 6 of the Works Contact: List of Statutory Compliances; and
xxxiv. Any statutory modifications or re-enactments thereto.

The contractor shall display abstract of prominent safety and labour laws at conspicuous places at site.

33. Penalty Against Safety Violation and Accident / Incident

The contractor shall comply with all the requirements laid down in the contract agreement-safety rules and regulation and BOCW Act 1996 and MBOCW Rules 2007, and any other safety & health requirements as a matter of general judiciousness. The safety violations leading to any type of incident and injury are not acceptable to the company and this will attract heavy penalty immediately at the discretion of Project Head or Safety Head of company including termination of the contract or removal of defaulting employees. The company project head shall communicate its decision to defaulting contractor officially and proper record shall be maintained. The penalty will be deducted from contractor's next running bill.

The nature of violations and the penalties are more particularly described below:

1. **For Category Level 1,2 & 3**
   a. First Safety Violation: Warning
   b. Second Safety Violation: Penalty as per schedule
   c. Third Safety Violation: Double penalty
   d. Fourth Safety Violation: Dismissal from site
2. **For Level 4 Category**
   a. First Incident: Penalty
   b. Second Incident: Dismissal

	Nature of Safety and Housekeeping Violations	**Penalty in Rs. including applicable GST (per violation)**
Level-1 Penalty		
1	Mandatory PPE's not used by employees of Contractor or not provided by Contractor (Safety helmet, safety shoes and reflective jacket).	Rs. 500/-
2	Other job specific PPE (hand gloves, safety goggles, welder goggles, dust mask, ear plugs etc.) not provided or used by employees of Contractor.	Rs. 500/-

3	Smoking in site premises (fire, explosion & health hazard)	Rs. 500/-
Level-2 Penalty		
4	Contractor's worker or staff working without mandatory safety induction (site orientation on hazards & incident).	Rs. 2000/-
5	Haphazard/ obstructive material stacking at work place, passages and in yards and fall of unsecured tools/ objects (trip, fall or struck-by hazard)	Rs. 2000/-
6	Inadequate up-keep at site and labour camp, unhygienic toilets, bath area, drinking water, fire/explosion hazard etc (health hazard & fire hazard)	Rs. 2000/-
7	Use of unsafe & unguarded power hand tools such as cutter, grinder etc. (Amputation risk)	Rs. 2000/-
8	Use of unfit vehicles (reverse horn, mirror, cabin gate, flood lights not provided, daily inspection missing) (vehicular incident).	Rs. 2000/-
9	Electric cables trailing on walkways, routed on floor, metallic structures, water (tripping hazard/ electrocution hazard).	Rs. 2000/-
10	Unsafe electric connections- damaged cables, multiple-joints, bare wire connection without 3-pin-plugs, multiple connections from single socket etc. (electrocutions, short circuit/ fire hazard)	Rs. 2000/-
11	Improper storage and unsecured handling of gas cylinders without wheel mounted trolley (toppling/ roll-down of cylinders, fire & explosion hazard)	Rs. 2000/-
Level-3 Penalty		
12	First aid kit not available with contractor (delayed medical assistance)	Rs. 10,000/-
13	Working at height without safety belt or safety belt not anchored (fall hazard/ fatality risk)	Rs. 10,000/-
14	Working at external face of towers and shafts without lifeline & fall grab arrester (fall hazard/ fatality risk)	Rs. 10,000/-
15	Poor housekeeping, debris, combustibles, excess construction accessories etc. not removed from work place (trip/ fall hazard or fire hazard)	Rs. 10,000/-
16	Debris/ construction material openly thrown from height, towers (bodily injury to several people)	Rs. 10,000/-

17	Hot work without fire extinguisher, fire blanket and permit or de-pressurised fire extinguisher (fire hazard/delay in fire mitigation)	Rs. 10,000/-
18	Safety catch net, fall protection barricading of trenches, excavation, shafts, MEP openings, slab edge not provided or removed without permission of Project Head/Safety Manager (fall of material/ person hazard).	Rs. 10,000/-
19	Unsafe working platform, scaffold, shuttering etc. (fall of material/ person hazard)	Rs. 10,000/-
20	Electric DB without 30mA ELCB/RCCB, non-standard design and earthing connection (electrocution hazard)	Rs. 10,000/-
21	Passenger hoist operation without competent operator, unsecured landing gates & lacking etc (fall hazard/crushing risk)	Rs. 10,000/-
22	Using material lifting tools, tackles, hoists, gondola & cranes without TPI (risk of machine failure/ fall of material/ incident)	Rs. 10,000/-
22	Operating earth moving vehicles without licensed & competent driver (un-authorised driving/ risk of vehicular incident)	Rs. 10,000/-
24	Violation to BOCW Requirements (legal threat to company, risk of incident)	Rs. 10,000/-
25	Deviation to work method statement and risk assessment (risk of incident & quality issue)	Rs. 10,000/-
26	Safety officer or safety steward not deployed as per contract condition (lack of safety management and supervision)	Rs. 10,000/-
27	Not complying with weekly safety observations, third party audits etc. on time and evidence of repetitive violations.	Rs. 10,000/-
28	Not attending project safety committee meeting (delay in decision & communication)	Rs. 10,000/-
29	Not reporting near-miss or incident within 24 hrs to company project head or safety manager (Deviation to regulation & delay in corrective/ preventive action).	Rs. 10,000/-
30	Any other safety violation not listed above, the Project Head shall levy penalty not amounting to more than Rs.10,000/-	Rs. 10,000/-
Level-4 Penalty		
31	Fire incident	Rs. 50,000/-
32	Leakage / spillage of flammable or toxic gas & liquids	Rs. 50,000/-

33	Medical treatment case (MT) due to negligence	Rs. 50,000/-
34	Dangerous Occurrence with grave potential as defined in BOCW Rules	Rs. 50,000/-
35	Lost time reportable incident (LTI)	Rs. 1,00,000/-
36	Fatality/ disability	Heavy Penalty and or disciplinary action as deemed fit basis investigation.

34. Additional Safety Rules & Regulations

The Company reserves the right to issue additional safety rules and regulations, standard operating procedures, work instructions, guidelines and checklists as and when required particularly under exceptional hazardous working conditions, new plant & machineries, change of operational conditions/ methods, construction of high-rise towers, location & neighbouring constraints of construction site etc and same shall be binding on the contractor to adhere and ensure additional safety precautions for safety of people and property at his time and cost.

35. Abbreviations

BOCW Act: The Building and Other Construction Workers (R E & C S) Act 1996.
DB: Distribution board
ELCB: Earth leakage circuit breaker
LPG: Liquified petroleum gas
LOTO: Lock-out and Tag-Out
LTI: Lost time reportable incident
MBOCW Rules: Maharashtra Building and Other Construction Workers (Regulation of Employment and Conditions of Service) Rules 2007.
MT: Medical treatment injury
MEP: Mechanical, electrical and plumbing
PPE: Personal protective equipment
PTW: Permit to work
RCCB: Residual current circuit breaker
SMART: Specific, measurable, achievable, realistic and time-bound
TBT: Tool box talk

36. Acknowledgement by Contractor

ACKNOWLEDGEMENT OF SAFETY RULES AND REGULATIONS BY CONTRACTOR

I, _____ (Name) _____(Designation) of _____(Company) being a Contractor for the project at _____, do hereby confirm that I have been briefed by the Project Head and Safety Manager on Safety Rules & Regulations. I fully understand the requirements and agree to abide by the same while working at project and shall ensure safety of workers. Any contravention of the rules and regulations and incident at project on our part shall render us fully accountable to all the consequences.

I confirm that I am a Director/ Authorised and appointed representative of the Company.

My contract value for this project is Rs. _____Crores / Lakhs. The Contract period is from _____to _____.

_

_____	_____
Signed by:	Witnessed by:
Contractor Project Manager / Owner.	Company / Project Head
Name: _____	Name: _____
Designation: _____	Designation: _____
Contact No: _____	Contact No: _____

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